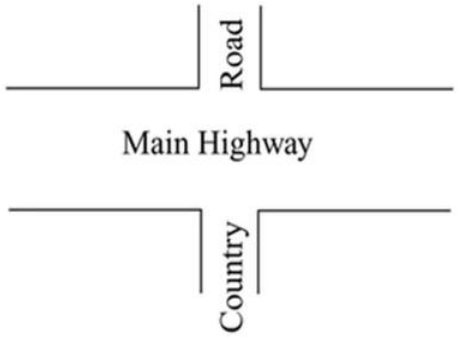
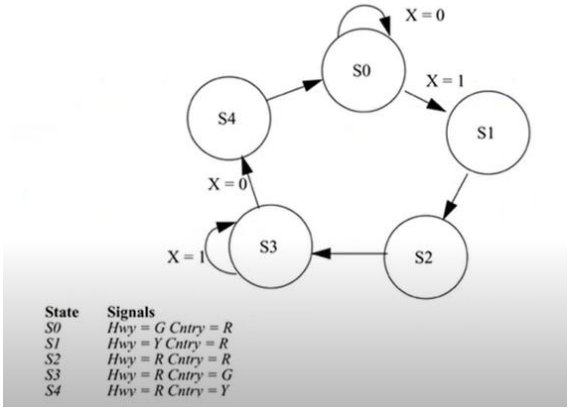
Traffic

Diagram:



Your default cars cross the highway and when detect a car x=1 in the country code will change the traffic colors and return again to normal when x=0.



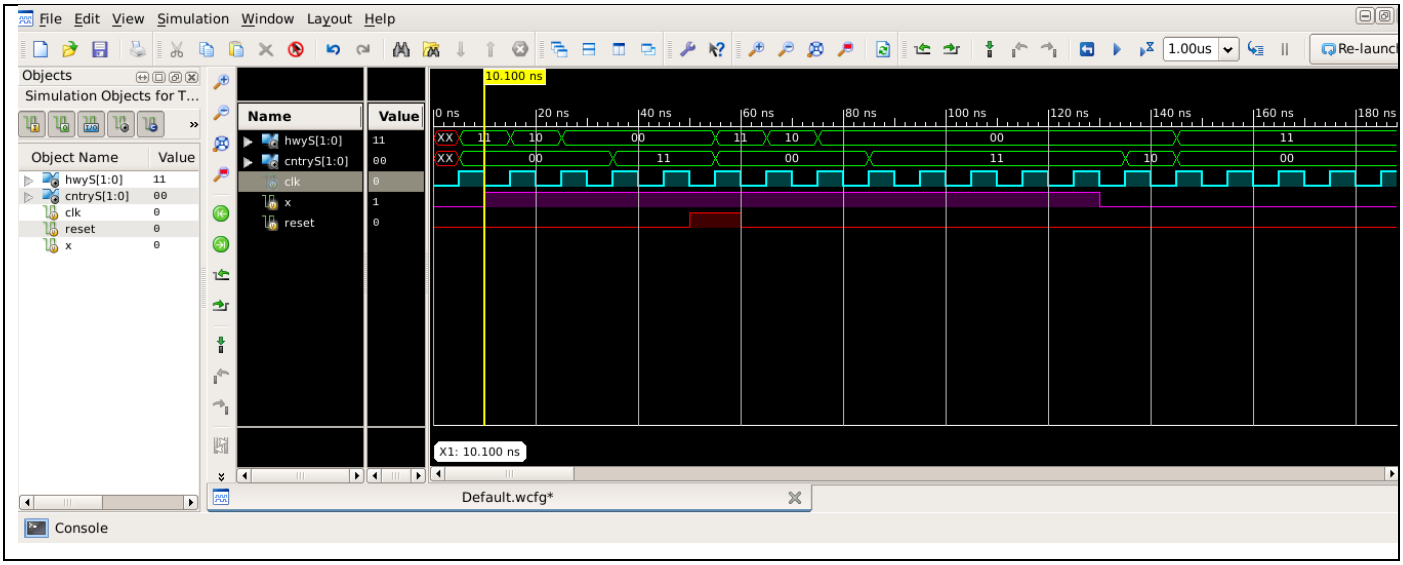
Verilog Code:

```
module traffic_control_signal (input x, clk, reset, output reg [1:0] hwyS, cntryS);
parameter green = 2'b11, yellow = 2'b10, red = 2'b00;
parameter s0 = 3'd0, s1 = 3'd1, s2 = 3'd2, s3 = 3'd3, s4 = 3'd4;
reg [2:0] state, next_state;

always @ (posedge clk)
if (reset) state <= s0;
else state <= next_state;

always @ (state)
case(state)
s0: begin hwyS = green; cntryS = red; end
s1: begin hwyS = yellow; cntryS = red;  end
s2: begin hwyS = red; cntryS = red;  end
s3: begin hwyS = red; cntryS = green; end
s4: begin hwyS = red; cntryS = yellow; end
default: begin hwyS = green; cntryS = red; end
endcase

always @ (state or x)
case (state)
s0: if(x) next_state = s1; else next_state = s0;
s1: next_state = s2;
s2: next_state = s3;
s3: if(x) next_state = s3; else next_state = s4;
s4: next_state = s0;
default: next_state = s0;
endcase

endmodule
```

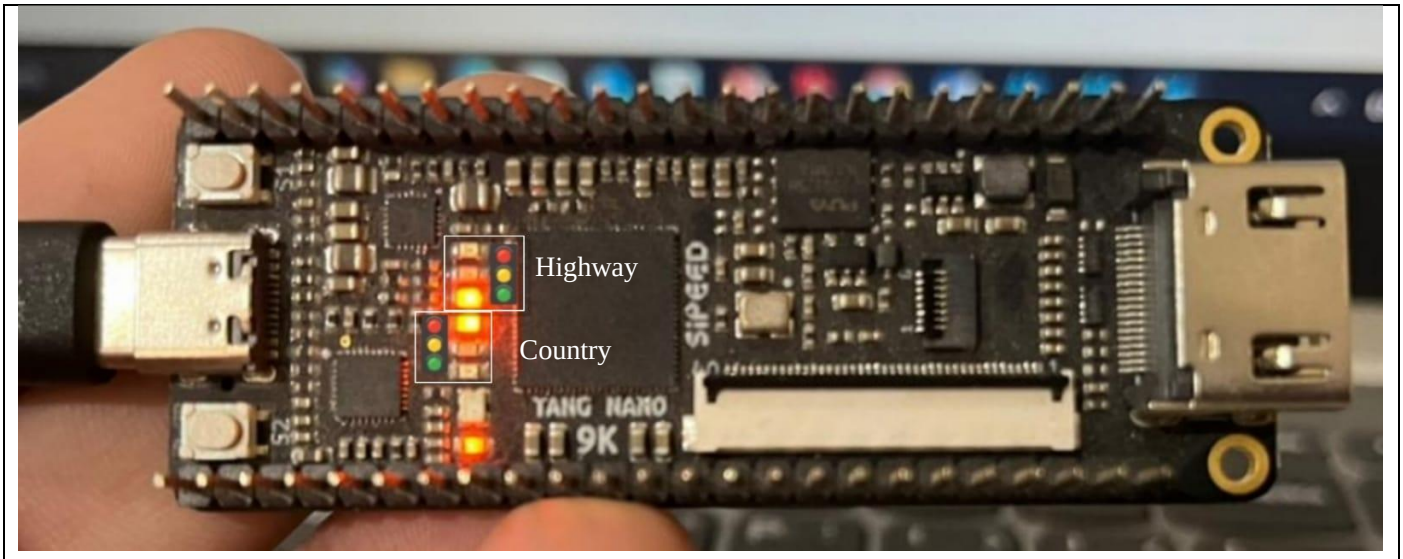
Verilog Code tb:

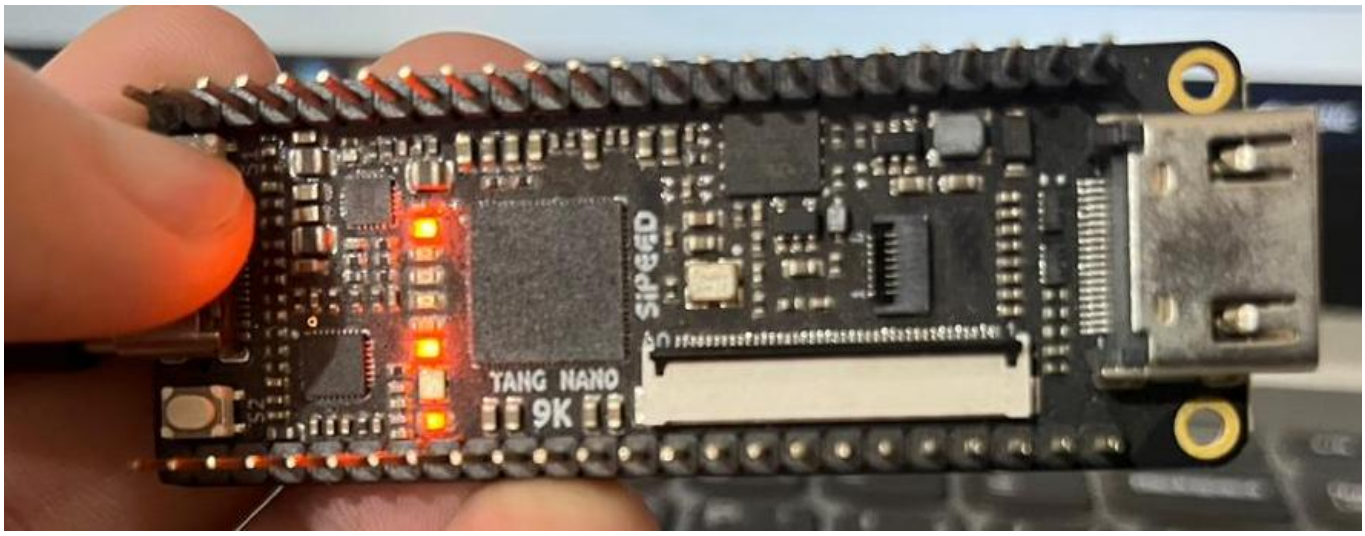
```
module Traffic_tb;
reg clk;
reg reset;
reg x;

wire [1:0] hwyS;
wire [1:0] cntryS;

Traffic uut (
.clk(clk),
.reset(reset),
.x(x),
.hwyS(hwyS),
.cntryS(cntryS)
);

initial
clk = 0;
always #5 clk = ~clk;
initial begin
x = 0;
reset = 0;
#10 x = 1;
#40 reset = 1;
#10 reset = 0;
#70 x = 0;
end

endmodule
```

Simulation:



FPGA:





Explanation:

we have 6 leds to implement every traffic light to each road, the first 3 leds for highway road and the second leds below for the country road. (1st led for each part is red then yellow and last one is the green)

at the beginning: the normal state is (S0):

green "the 3rd led to highway road" and red "the 4th led for country road"

then when X button is pressed the states will settle on (S3):

red "the 1st led to highway road" and green "the 6th led to country road"

passing through the states (S1 & S2).